

**MATH 415/515 Advanced Linear Algebra**  
**Exam 1 Review**

## **1 List of Topics**

1. Review

- (a) Vector space,  $V$
- (b) Subspace,  $W$
- (c) Linear combination, span, spanning set
- (d) Linear independence
- (e) Basis
- (f) Dimension
- (g) Eigenvalue, eigenvector, eigenspace, algebraic multiplicity, geometric multiplicity
- (h) Null space,  $\text{NS}(A)$ , column space,  $\text{CS}(A)$ , row space,  $\text{RS}(A)$

2. Inner product on  $\mathbb{R}^n$ , dot product

- (a) Length, norm,  $\|\mathbf{v}\|$
- (b) Unit vector,  $\hat{\mathbf{v}}$
- (c) Distance,  $\|\mathbf{v} - \mathbf{u}\|$
- (d) Angle,  $\theta$
- (e) Orthogonality
  - i. Orthogonal complement,  $W^\perp$
  - ii. Orthogonal set, orthogonal basis
  - iii. Orthogonal projection of  $\mathbf{x}$  onto a subspace  $W$ ,  $\mathbf{x}' = \text{proj}_W(\mathbf{x})$
  - iv. The component of  $\mathbf{x}$  orthogonal to  $W$ ,  $\mathbf{z}$
  - v. Orthonormal set, orthonormal basis
  - vi. Orthogonal matrix,  $U$
  - vii. The Gram-Schmidt process
  - viii. QR-factorization
  - ix. Least-squares solution to  $A\mathbf{x} = \mathbf{b}$ , normal equations

## 2 Sample Questions

Short-answer questions for Exam 1 may come from the following list. If I think of additional problems, I reserve the right to include them on Exam 1.

### Review

1. Define a subspace  $W$  of a vector space  $V$ .
2. Provide three examples of vector spaces, none of which are  $\mathbb{R}^n$ .
3. Define a linear combination of the vectors  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ .
4. Define  $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ .
5. Define linear independence for  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ .
6. Define a basis of a subspace  $W$ .
7. State the theorem from class that lists four equivalent statements for the span of a collection of vectors.
8. Define the dimension of a subspace  $W$ . Include definitions for both a finite-dimensional subspace and an infinite-dimensional subspace.
9. Provide an example of a finite-dimensional vector space (not  $\mathbb{R}^n$ ) and an infinite-dimensional vector space.
10. State the Basis Theorem.
11. Define an eigenvalue,  $\lambda$ , and an associated eigenvector,  $\mathbf{v}$ , of a matrix  $A$ .
12. Define the algebraic multiplicity,  $a_\lambda$ , and geometric multiplicity,  $g_\lambda$  for an eigenvalue  $\lambda$ .
13. State a theorem relating the concepts of eigenvectors and linear independence.
14. Define the row space,  $\text{RS}(A)$ , column space,  $\text{CS}(A)$ , and null space,  $\text{NS}(A)$ , for an  $m \times n$  matrix.
15. State the definition of a linear transform,  $T : V \rightarrow W$ .
16. Define similar matrices,  $A$  and  $B$ .

### Inner product

17. Define the inner product on  $\mathbb{R}^n$ . Include the standard definition, as well as the definition that uses matrix multiplication.
18. Define the length or norm of a vector.
19. Define orthogonality between two vectors.
20. State the Pythagorean Theorem.
21. Define the orthogonal complement of a subspace  $W$ .
22. State a theorem that relates the orthogonal complements of  $\text{NS}(A)$ ,  $\text{CS}(A)$ , and  $\text{RS}(A)$ .
23. How can the inner product be used to define the angle between two vectors?

### Orthogonal sets

24. Define an orthogonal basis for a subspace  $W$ .
25. State a theorem that shows how to calculate a linear combination for an orthogonal basis.

26. Define the orthogonal projection of  $\mathbf{x}$  onto a subspace  $W$ ,  $\text{proj}_W(\mathbf{x})$ .
27. Define an orthonormal set,  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ .
28. Define an orthonormal basis for a subspace  $W$ .
29. State an if-and-only-if theorem for matrices with orthonormal columns.
30. Define an orthogonal matrix,  $U$ .

### Orthogonal projections

31. State the Orthogonal Decomposition Theorem.
32. State the Best Approximation Theorem.
33. State a theorem that describes how to calculate an orthogonal projection onto a subspace  $W$  with an orthonormal basis of  $W$ .
34. Given an  $m \times n$  matrix  $U$  with orthonormal columns, explain the results of the computations  $UU^T$  versus  $U^TU$ . Are they ever the same?
35. Between class and the homework we have dealt with both matrices with orthonormal columns and orthogonal matrices. Are they the same thing? (This means that  $U$  is an orthogonal matrix if and only if  $U$  has orthonormal columns.) If they are the same, prove the if-and-only-if statement. If they are not the same, what is different between the two types of matrices? (Meaning, one of them should have at least one property that the other one does not. Provide a list of properties of both types of matrices and identify those items that are not on both lists.)

### The Gram-Schmidt process

36. Explain the purpose of the Gram-Schmidt process. (What do we need before we can start the process and what does it produce?)
37. How does the Gram-Schmidt process work? (What happens at each stage of the process? A picture might help.)
38. State the QR-Factorization Theorem.
39. How do we calculate a QR-factorization of  $A$ ?

### Least-squares problems

40. Define a least-squares solution of  $A\mathbf{x} = \mathbf{b}$ .
41. Define the normal equations for  $A\mathbf{x} = \mathbf{b}$ .
42. How do we calculate the error associated with a least-squares solution?

### 3 Practice Problems

1. Either provide a proof or counterexample as the justification for each statement.
  - (a) (T/F) If  $c$  is any scalar, then  $\|c\mathbf{v}\| = c\|\mathbf{v}\|$ .
  - (b) (T/F) If  $\mathbf{x}$  is orthogonal to both  $\mathbf{u}$  and  $\mathbf{v}$ , then  $\mathbf{x}$  must be orthogonal to  $\mathbf{u} - \mathbf{v}$ .
  - (c) (T/F) If  $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ , then  $\mathbf{u}$  and  $\mathbf{v}$  are orthogonal.
  - (d) (T/F) If  $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ , then  $\mathbf{u}$  and  $\mathbf{v}$  are orthogonal.
  - (e) (T/F) The orthogonal projection of  $\mathbf{x}$  onto  $\mathbf{u}$  is a scalar multiple of  $\mathbf{x}$ .
  - (f) (T/F) If a vector  $\mathbf{x}$  coincides with its orthogonal projection onto a subspace  $W$ , then  $\mathbf{x} \in W$ .
  - (g) (T/F) If  $W$  is a subspace of  $\mathbb{R}^n$ , then  $W$  and  $W^\perp$  have no vectors in common.
  - (h) (T/F) If a matrix  $U$  has orthonormal columns, then  $UU^T = I$ .
  - (i) (T/F) A square matrix with orthogonal columns is an orthogonal matrix.
  - (j) (T/F) If a square matrix has orthonormal columns, then it also has orthonormal rows.
2. Prove that if an  $n \times n$  matrix  $U$  satisfies  $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ , for all  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ , then  $U$  is an orthogonal matrix. Hint: this holds for ANY vectors in  $\mathbb{R}^n$ . Consider the standard basis  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  and consider what  $U\mathbf{e}_i$  computes, for  $i = 1, \dots, n$ .
3. Let  $U$  be an orthogonal matrix.
  - (a) Prove that  $\det(U) = \pm 1$ . Hint: recall that  $\det(A^T) = \det(A)$ .
  - (b) Prove that for any eigenvalue of  $U$ ,  $|\lambda| = \pm 1$ . Hint: recall that  $\|U\mathbf{x}\| = \|\mathbf{x}\|$ .
4. A *Householder matrix* has the form  $Q = I - 2\mathbf{u}\mathbf{u}^T$ , where  $\mathbf{u}$  is a unit vector. Prove that  $Q$  is an orthogonal matrix. Hint: compute  $Q^T Q$ .
5. Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be a linear transformation that preserves norms,  $\|T(\mathbf{x})\| = \|\mathbf{x}\|$ , for all  $\mathbf{x} \in \mathbb{R}^n$ . Prove that  $T$  also preserves orthogonality, or if  $\mathbf{x} \cdot \mathbf{y} = 0$ , then  $T(\mathbf{x}) \cdot T(\mathbf{y}) = 0$ . Hint: use the Pythagorean Theorem.

**Theorem** (The Invertible Matrix Theorem). Let  $A$  be an  $n \times n$  matrix. The following are equivalent.

- (1)  $A$  is an invertible matrix.
- (2)  $A$  is row equivalent to the  $n \times n$  identity matrix.
- (3)  $A$  has  $n$  pivot rows.
- (4) The equation  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution.
- (5) The columns of  $A$  form a linearly independent set.
- (6) The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  is one-to-one/injective.
- (7) The equation  $A\mathbf{x} = \mathbf{b}$  has at least one solution for each  $\mathbf{b} \in \mathbb{R}^n$ .
- (8) The columns of  $A$  span  $\mathbb{R}^n$ .
- (9) The linear transformation  $x \mapsto A\mathbf{x}$  is onto/surjective.
- (10) There is an  $n \times n$  matrix  $C$  such that  $CA = I$ .
- (11) There is an  $n \times n$  matrix  $D$  such that  $AD = I$ .
- (12)  $A^T$  is an invertible matrix.
- (13)  $\det(A) \neq 0$ .
- (14) The columns of  $A$  form a basis of  $\mathbb{R}^n$ .
- (15)  $\text{CS}(A) = \mathbb{R}^n$ .
- (16)  $\text{rk}(A) = n$ .
- (17)  $\nu(A) = 0$ .
- (18)  $\text{NS}(A) = \{\mathbf{0}\}$ .
- (19) The number 0 is not an eigenvalue of  $A$ .